

# MUHAMMAD AWAIS

Senior Software Engineer

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## PROFESSIONAL SUMMARY

Senior Software Engineer with **3+ years** of experience building **scalable web applications, cloud-native solutions**, and **AI-driven platforms**. Expertise in MEAN/MERN stack, cloud deployments, and real-time communication systems. Adept at leading end-to-end development, integrating third-party APIs, and delivering high-impact software products.

## TECHNICAL SKILLS

| Domain             | Technologies                                                                  |
|--------------------|-------------------------------------------------------------------------------|
| Languages          | JavaScript, TypeScript, Python                                                |
| Frontend           | React, Angular, Tailwind CSS, Bootstrap, MUI                                  |
| Backend            | Node.js, Express.js, REST APIs, Socket.IO                                     |
| Databases          | MongoDB, Firebase, Redis                                                      |
| AI & Communication | OpenAI LLM, Twilio, Deepgram STT/TTS, Google STT/TTS, Infobip                 |
| Cloud & DevOps     | AWS (EC2, S3, CloudFront), Google Cloud Run, Docker, Kubernetes, Nginx, CI/CD |
| Tools & Practices  | Git, Bitbucket, Jest, Postman, Jira, Agile/Scrum                              |

## PROFESSIONAL EXPERIENCE

Senior Software Engineer — Callvio

📅 May 2025 – Present | 🔗 [callvio.co](#) • [Dashboard](#)

Led development of an AI-powered Receptionist App for automating business customer interactions through intelligent voice assistance.

- Architected a **full-stack AI receptionist platform** handling live customer calls with natural language understanding and automated responses
- Engineered a responsive **React frontend** with Firebase integration, achieving **sub-second load times** deployed on Vercel
- Designed scalable **Node.js microservices** integrating **Twilio** for voice routing, **Deepgram & Google STT/TTS** for real-time speech processing, and **OpenAI LLM** for conversation intelligence
- Implemented **Infobip WebSocket** integration for real-time caller speech capture and Jazz telephony support

- Built intelligent **appointment booking**, business FAQ handling, and real-time automated response systems
- Set up **CI/CD pipelines** with automated testing; deployed backend on **Google Cloud Run** for auto-scaling
- Integrated **Redis** for session management, caching, and real-time data processing ensuring low-latency responses
- Combined telephony, AI, and cloud-native technologies, significantly improving **customer engagement and operational efficiency**

**Tech:** React Node.js Firebase Redis Twilio OpenAI Deepgram Google TTS/STT Infobip Cloud Run Vercel

## Senior Software Engineer — Binary Tech (DoctorNow)

 May 2022 – April 2025 |  [docnow.ai](https://docnow.ai) • [Live Dashboard](#)

*Developed and maintained core features for DoctorNow — a comprehensive telemedicine platform with EHR, real-time consultations, and digital prescriptions.*

- Led **front-end architecture** using Angular with lazy loading, AOT compilation, and optimized bundle sizes for desktop and mobile
- Engineered scalable **backend APIs** using Node.js and Express with modular architecture, middleware pipelines, and comprehensive error handling
- Designed and optimized **MongoDB schemas** with indexing, aggregation pipelines, and query optimization for managing large volumes of patient data
- Built **real-time chat** modules using Socket.IO for doctor-patient communication during consultations
- Implemented secure **authentication flows** using JWT, refresh tokens, HTTPS enforcement, rate limiting, and input validation/sanitization
- Integrated **Twilio** (SMS/voice), **Zoom SDK** (video consultations), and **payment gateways** for seamless third-party services
- Developed **digital prescriptions**, video consultation rooms, appointment scheduling, and patient history tracking
- **Reduced page load time by 40%** and **increased user engagement by 25%** through code optimization, CDN integration, and UI revamp
- Led **Agile ceremonies** — sprint planning, code reviews, mentoring junior developers, and CI/CD pipeline management

**Tech:** Angular Node.js Express.js MongoDB Socket.IO JWT Twilio Zoom SDK AWS Docker Jest



## EDUCATION

**Bachelor of Science in Computer Science** | 2018 – 2022 *Punjab University College of Information Technology (PUCIT), Lahore, Pakistan*